MATH-18A College Algebra **Exam 3 Practice**

The exam will have around 6 questions, and one optional bonus question on the last page. You are allowed a non-graphing calculator and ONE 3×5 notecard. **Show all your work** and **box your answers**.

Section 3.1

1. (4 points) Let $f(x) = x^3 - 4$. **Evaluate** and simplify $f(t+2)$.

For more practice, repeat with the following:

2. $f(-2) =$

3. $2 \cdot f(3) =$

4. $f(t) =$

5. $f(t-4) =$

Let $g(x) = \dfrac{2x+7}{x-1}$.

6. $g(0) =$

7. $g(1) =$

8. $-3 \cdot g(k) =$

9. $g(k+1) =$

Section 3.2

10. (6 points) Write the **domain** of $f(x) = \dfrac{4-x}{(2x+3)(x-7)}$ using interval notation.

For more practice, repeat with the following:

11. $f(x) = \dfrac{7}{3x(x-1)}$

12. $f(x) = \dfrac{x}{10+3}$

13. $f(x) = \dfrac{3x+1}{x-11}$

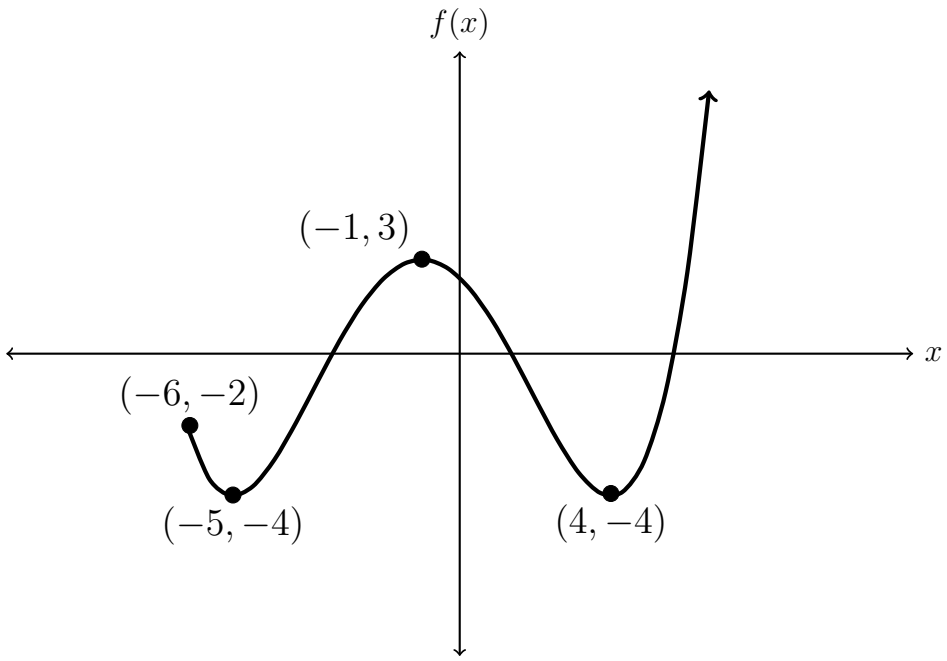
14. $f(x) = \dfrac{x^2}{x^2-1}$

15. Challenge: $f(x) = \sqrt{-x}$

16. Challenge: $f(x) = \dfrac{1}{\sqrt{x+5}}$

Section 3.3

17. (6 points) For the following questions, refer to the graph of $f(x)$ shown below.



a) On what interval(s) of x is $f(x)$ **increasing**? Use interval notation.

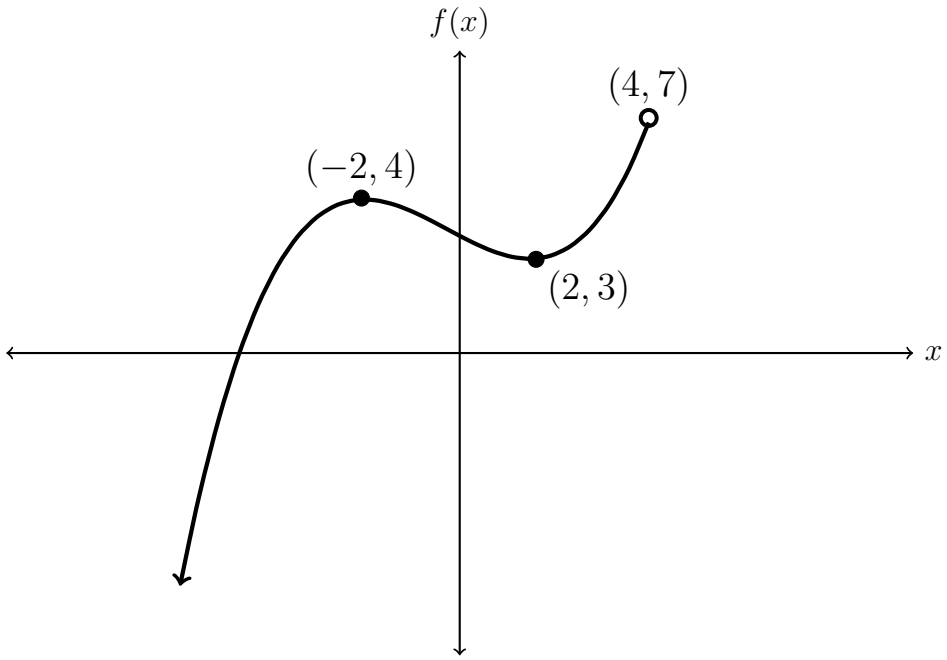
b) On what interval(s) of x is $f(x)$ **decreasing**? Use interval notation.

c) Write the coordinate(s) of the **local maximum(s)**.

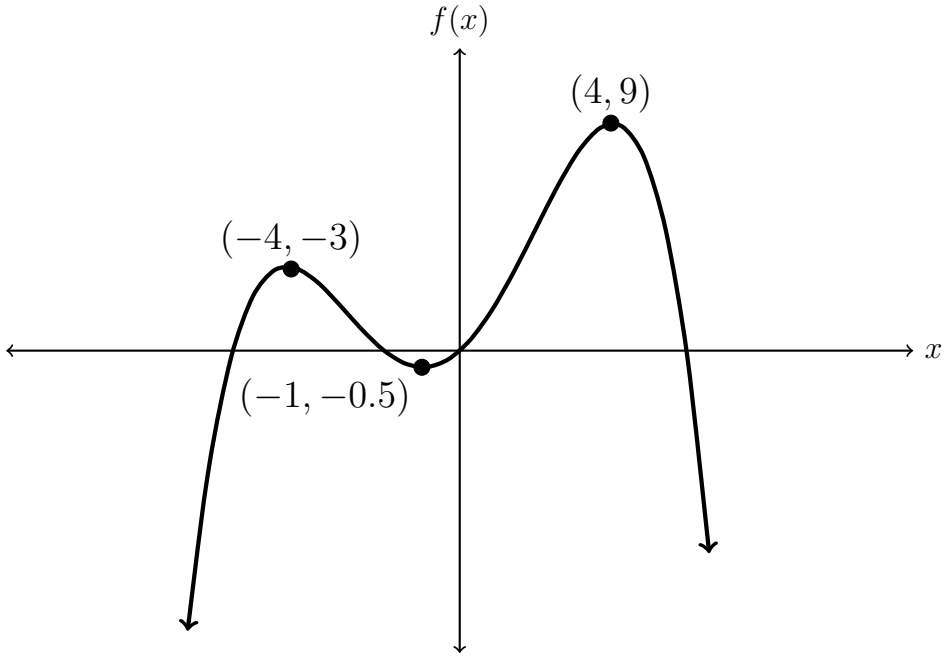
d) Write the coordinate(s) of the **local minimum(s)**.

More more practice, repeat with the following:

18. Use the graph...

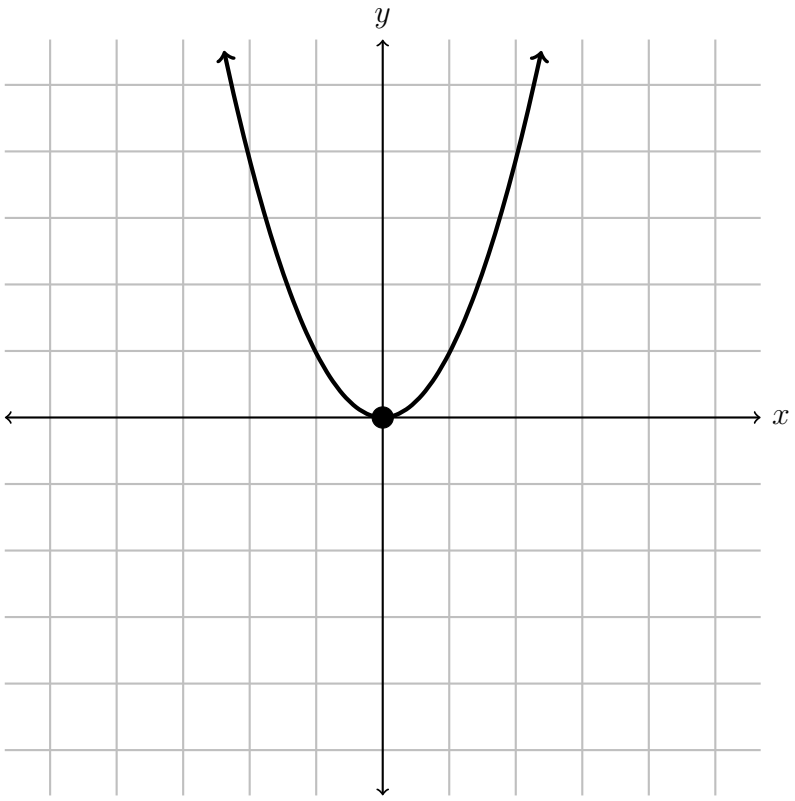


19. Use the graph...



Section 3.4

20. (4 points) **Identify** a, h, and k in the transformed function $f(x-4)+3$ and briefly explain in words what they do. Then, using the graph of $f(x)$ below, **draw** the transformed function on the same axes.



More more practice, repeat with the following:

21. parent function: $f(x)=\sqrt{x}$; transformed function: $-f(x+2)+3$

22. parent function: $f(x)=|x|$; transformed function: $2f(x-1)+7$

23. parent function: $f(x)=x^3$; transformed function: $f(x+11)-1$

Section 3.5

24. (6 points) Let $f(x) = \sqrt{x-5}$ and $g(x) = 7(x+4)$. [On the exam, you will only be asked to solve TWO specified combinations, not all six.]

a) Solve $(f+g)(x)$.

b) Solve $(f-g)(x)$.

c) Solve $(g-f)(x)$.

d) Solve $(fg)(x)$.

e) Solve $\left(\frac{f}{g}\right)(x)$.

f) Solve $\left(\frac{g}{f}\right)(x)$

More more practice, repeat with the following:

25. $f(x) = -x^5 + 4$ and $g(x) = \frac{x}{7}$

26. $f(x) = \frac{1}{x-3}$ and $g(x) = 2(x^2+1)$

27. $f(x) = |x| + 3$ and $g(x) = (x-3)^2$

Section 3.6

28. (6 points) Find the **inverse** of $f(x) = 3(x+1)^3$ and **check** your answer.

Check $(f \circ f^{-1})(x)$:

Check $(f^{-1} \circ f)(x)$:

More more practice, repeat with the following:

29. $f(x) = \dfrac{7}{x^2+2}$

30. $f(x) = \dfrac{3}{x} + 9$

31. $f(x) = \sqrt[5]{2x-1}$

32. Challenge: $f(x) = x^2 - 6$